

6CS030: Big Data

Workshop - 8

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SECTION 1

1. What is ELK stack? Explain it using its architecture.
2. What is Shard in ElasticSearch?
3. What are the three major components of Logstash?
4. Perform a case study on the possible usage of ELK stack by any one of the following companies leveraging big data.
 - a. Netflix
 - b. LinkedIn
 - c. Medium
5. Explore Fuzzy Search in ELK.

SECTION 2

For this, you need to install the ELK stack in your machine.

Elastic Running:

```
[narayan@narayan ~]$ % /opt/homebrew/opt/elasticsearch-full/bin/elasticsearch
warning: usage of JAVA_HOME is deprecated, use ES_JAVA_HOME
warning: usage of JAVA_HOME is deprecated, use ES_JAVA_HOME
Apr 24, 2026 7:28:51 AM sun.util.locale.provider.LocaleProviderAdapter <clinit>
WARNING: COMPAT locale provider will be removed in a future release
[2026-04-24T07:28:52.496]INFO [l.o.e.n.Node] [narayan.local] version[7.17.4], pid[51421], build[default/tar/79878662c54c886ae89286c685d9f1051a9d6411/2022-05-18T18:04:20.964345128Z], OS[Mac
OS X/26.4.1/aarch64], JVM[Homebrew/OpenJDK 64-Bit Server VM/21.0.10/21.0.10]
[2026-04-24T07:28:52.496]INFO [l.o.e.n.Node] [narayan.local] JVM home [/opt/homebrew/Cellar/openjdk@21.0.10/libexec/openjdk.jdk/Contents/Home], using bundled JDK [false]
[2026-04-24T07:28:52.496]INFO [l.o.e.n.Node] [narayan.local] JVM arguments [-Xshare:auto, -Des.networkaddress.cache.ttl=60, -Des.networkaddress.cache.negative.ttl=10, -XX:AlwaysPreTouch, -
Xsasl, -Djava.awt.headless=true, -Dfile.encoding=UTF-8, -Djna.nosys=true, -Xi-DatStackTraceInFastThrow, -XX:ShowCodeDetailsInExceptionMessages, -Dio.netty.noUnsafe=true, -Dio.netty.noKeySetOptimization
=true, -Dio.netty.recycler.maxCapacityPerThread=0, -Dio.netty allocator.numDirectArenas=0, -Dlog4j.shutdownHookEnabled=false, -Dlog4j2.disable.jmx=true, -Dlog4j2.formatMsgNoLookups=true, -Djava.locale.pro
viders=SPI, COMPAT, --add-opens=java.base/java.io=ALL-UNNAMED, -Djava.security.manager=allow, -XX:-UseG1GC, -Djava.io.tmpdir=/var/folders/x3/7mv598bn3c7gnxp5jnxzprwh0808gn/T/elasticsearch-64437148687864628
37, -XX:HeapDumpOnOutOfMemoryError, -XX:ExitOnOutOfMemoryError, -XX:HeapDumpPath=data, -XX:ErrorFile=logs/hs_err_pid@%.log, -Xloggc*=gc-agent:trace,sa:heapint:file:/opt/homebrew/var/log/elasticsearch/gc.l
og;utctime,pid,tags:filecount%32,filesize%64m, -Xms8192m, -Xmx8192m, -XX:MaxDirectMemorySize=4294967296, -XX:InitiatingHeapOccupancyPercent=30, -XX:G1ReservePercent=25, -Des.path.home=/opt/homebrew/Cellar
/elasticsearch-full/7.17.4/libexec, -Des.path.conf=/opt/homebrew/etc/elasticsearch, -Des.distribution.flavor=default, -Des.distribution.type=tar, -Des.bundled_jdk=true]
[2026-04-24T07:28:53.384]INFO [l.o.e.p.PluginsService] [narayan.local] loaded module [analysis-common]
[2026-04-24T07:28:53.384]INFO [l.o.e.p.PluginsService] [narayan.local] loaded module [analysis-common]
```

Kibana Running:

```
[narayan@narayan ~]$ % /opt/homebrew/opt/kibana-full/bin/kibana
log [07:25:43.175] [info][plugins-service] Plugin "metricsEntities" is disabled.
log [07:25:43.222] [info][server][Preboot][http server running at http://localhost:5601]
log [07:25:43.244] [warning][config][deprecation] Environment variable "KIBANA_PATH_CONF" is deprecated. It has been replaced with "KBN_PATH_CONF" pointing to a config folder
log [07:25:43.261] [warning][config][deprecation] Environment variable "DATA_PATH" will be removed. It has been replaced with kibana.yml setting "path.data"
log [07:25:43.266] [warning][config][deprecation] Starting in 8.0, the Kibana logging format will be changing. This may affect you if you are doing any special handling of your Kibana logs, such as in
gesting logs into Elasticsearch for further analysis. If you are using the new logging configuration, you are already receiving logs in both old and new formats, and the old format will simply be going aw
ay. If you are not yet using the new logging configuration, the log format will change upon upgrade to 8.0. Beginning in 8.0, the format of JSON logs will be EC2-compatible JSON, and the default pattern l
og format will be configurable with our new logging system. Please refer to the documentation for more information about the new logging format.
log [07:25:43.267] [warning][config][deprecation] The default mechanism for Reporting privileges will work differently in future versions, which will affect the behavior of this cluster. Set "pack.re
porting.roles.enabled" to "false" to adopt the future behavior before upgrading.
log [07:25:43.268] [warning][config][deprecation] User sessions will automatically time out after 0 hours of inactivity starting in 8.0. Override this value to change the timeout.
log [07:25:43.269] [warning][config][deprecation] Users are automatically required to log in again after 30 days starting in 8.0. Override this value to change the timeout.
log [07:25:43.378] [info][plugins-system][startup] Setting up 1131 plugins: translation, license, globalSearch, globalSearchProviders, features, license@findcard-code, usageCollection, space@copy taskM
```

Logstash Running:

```
[narayan@narayan ~]$ % /opt/logstash/bin/logstash -f /opt/logstash/config/logstash-simple.conf
Using JAVA_HOME defined java: /opt/homebrew/Cellar/openjdk@21.0.10/libexec/openjdk.jdk/Contents/Home
WARNING: Using JAVA_HOME while Logstash distribution comes with a bundled JDK.
DEPRECATION: The use of JAVA_HOME is now deprecated and will be removed starting from 8.0. Please configure LS_JAVA_HOME instead.
Sending Logstash logs to /opt/logstash/logs which is now configured via log4j2.properties
[2026-04-24T07:48:26.836]INFO [logstash.runner] [log4j configuration path used is: /opt/logstash/config/log4j2.properties]
[2026-04-24T07:48:26.841]INFO [logstash.runner] [Starting Logstash ("logstash.version">"7.17.4", "jruby.version">"jruby 9.2.20.1 (2.5.8) 2021-11-30 2a2962fbd1 OpenJDK 64-Bit Server VM 21.0.10
on 21.0.10 <indy> jit [darwin-aarch64])"]
[2026-04-24T07:48:26.842]INFO [logstash.runner] [JVM bootstrap flags: [-Xms1g, -Xmx1g, -Djava.awt.headless=true, -Dfile.encoding=UTF-8, -Djdk.io.File.enableADS=true, -Djruby.compile.invokedynam
ic=true, -Djrubyjit.threshold=0, -Djruby.regexp.interruptible=true, -XX:-HeapDumpOnOutOfMemoryError, -Djava.security.egd=file:/dev/urandom, -Dlog4j2.isThreadContextMapInherited=true, --add-opens=java.b
ase/java.nio.channels=ALL-UNNAMED, --add-opens=java.base/java.io=ALL-UNNAMED]]
[2026-04-24T07:48:26.848]INFO [logstash.settings] [Creating directory (:setting=>"path.queue", :path=>"/opt/logstash/data/queue")]
[2026-04-24T07:48:26.851]INFO [logstash.settings] [Creating directory (:setting=>"path.dead_letter_queue", :path=>"/opt/logstash/data/dead_letter_queue")]
[2026-04-24T07:48:26.867]WARN [logstash.config.source.multilocal] Ignoring the 'pipelines.yml' file because modules or command line options are specified
[2026-04-24T07:48:26.873]INFO [logstash.agent] [No persistent UUID file found. Generating new UUID (:uuid=>"8897ab50-b7ce-4a3c-9f39-68f7b973cd86", :path=>"/opt/logstash/data/uuid")]
[2026-04-24T07:48:26.958]ERROR [logstash-instrument.periodicpoller.jvm] Unknown garbage collector name (:name=>"G1 Concurrent GC")
[2026-04-24T07:48:27.102]INFO [logstash.agent] [Successfully started Logstash API endpoint (:port=>5600, :url_enabled=>false)]
```

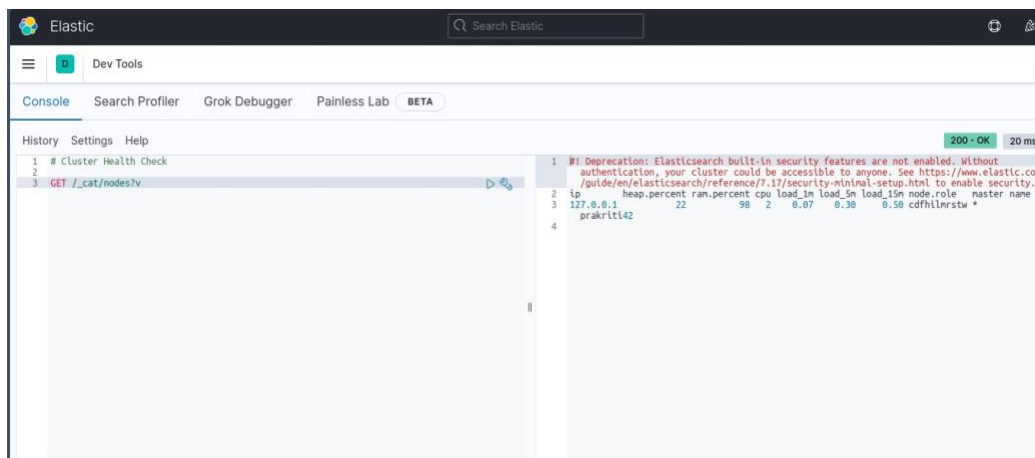
SECTION 3:

As part of building primary familiarity with ELK lets try exploring the provided elastic apis for data manipulation using the Kibana interface.

To start with this, open up Kibana on a web browser and go to

`http://localhost:5601/app/dev_tools#/console`

this should lead you to the Console that will allow you to kick start the task. The GUI looks like the following.



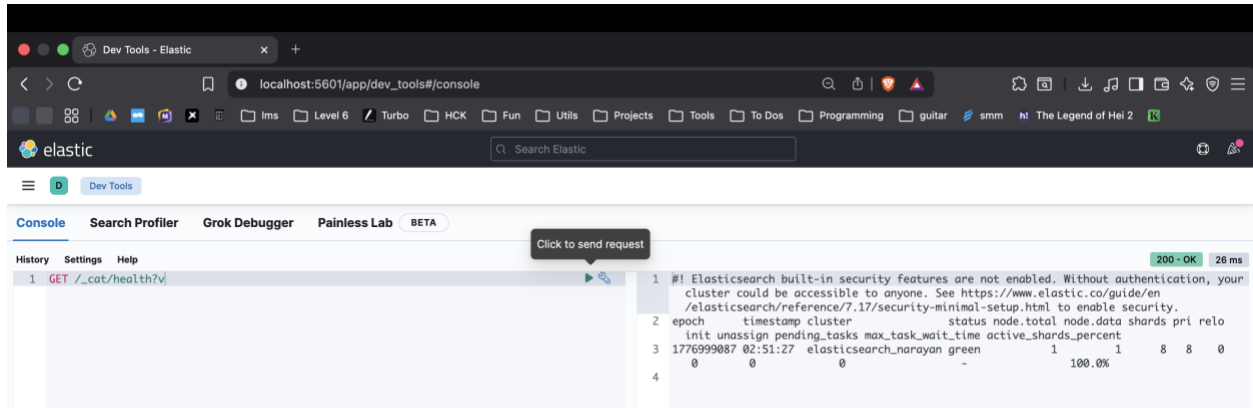
Make sure you have elasticsearch enabled too.

Checking Cluster Health

Your task first is to try out a way to make a cluster health check via an Elastic API call.

Consider trying the command

GET /_cat/health?v



Exercise To do

1. Observe and write down your understanding of the result. Also, try sending a call request without the ?v, note down the difference if present.

Making sense of the response:

Epoch

Field	Value	Description
epoch	1776999087	Unix timestamp, machine time format
timestamp	02:51:27	Human-readable time of the snapshot
cluster	elasticsearch_narayan	Name of the Elasticsearch cluster
status	green	Overall health status (green = fully healthy)
node.total	1	Total number of nodes in the cluster
node.data	1	Number of nodes that store data
shards	8	Total number of shards (primary + replicas)
pri	8	Number of primary shards (original data holders)

relo	0	Shards currently relocating between nodes
init	0	Shards currently initializing (starting up/recovering)
unassign	0	Shards not assigned to any node (problem if > 0)
pending_tasks	0	Number of cluster tasks waiting to be executed
max_task_wait_time	-	Longest wait time of pending tasks (“-” = none waiting)
active_shards_percent	100.0%	Percentage of shards that are active and working

The difference between having ‘v’ and not having ‘v’ at the end of request is to include the labels (“epoch”, ... , “active_shards_percent”) along with the values while displaying the health status results. Excluding ‘v’ displays the values only.

Including v:

```
1 #! Elasticsearch built-in security features are not enabled. Without authentication, your cluster could be accessible to anyone. See https://www.elastic.co
2 /guide/en/elasticsearch/reference/7.17/security-minimal-setup.html to enable security.
3 epoch timestamp cluster status node.total node.data shards pri relo init unassign pending_tasks max_task_wait_time active_shards_percent
4 1776999978 03:06:18 elasticsearch_narayan green 1 1 8 8 0 0 0 0 0 0 - 100.0%
```

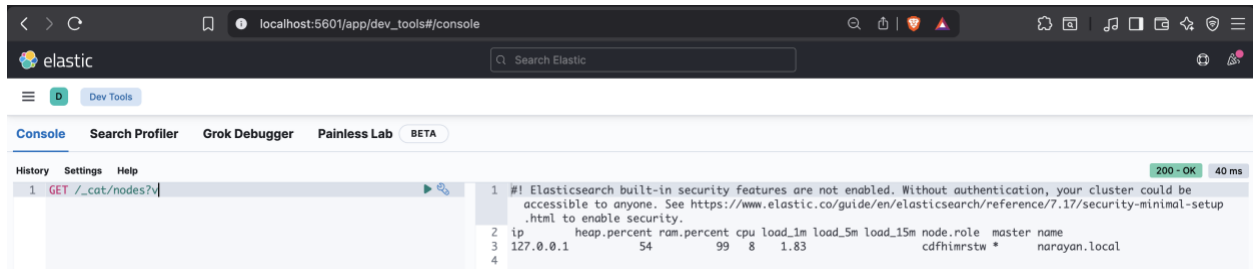
Excluding v:

```
1 #! Elasticsearch built-in security features are not enabled. Without authentication, your cluster could be accessible to anyone. See https://www.elastic.co
2 /elasticsearch/reference/7.17/security-minimal-setup.html to enable security.
3 17769999616 03:00:16 elasticsearch_narayan green 1 1 8 8 0 0 0 0 - 100.0%
```

2. Make an api call to check the node health.

Node health can be checked using following command:

GET /_cat/nodes?v

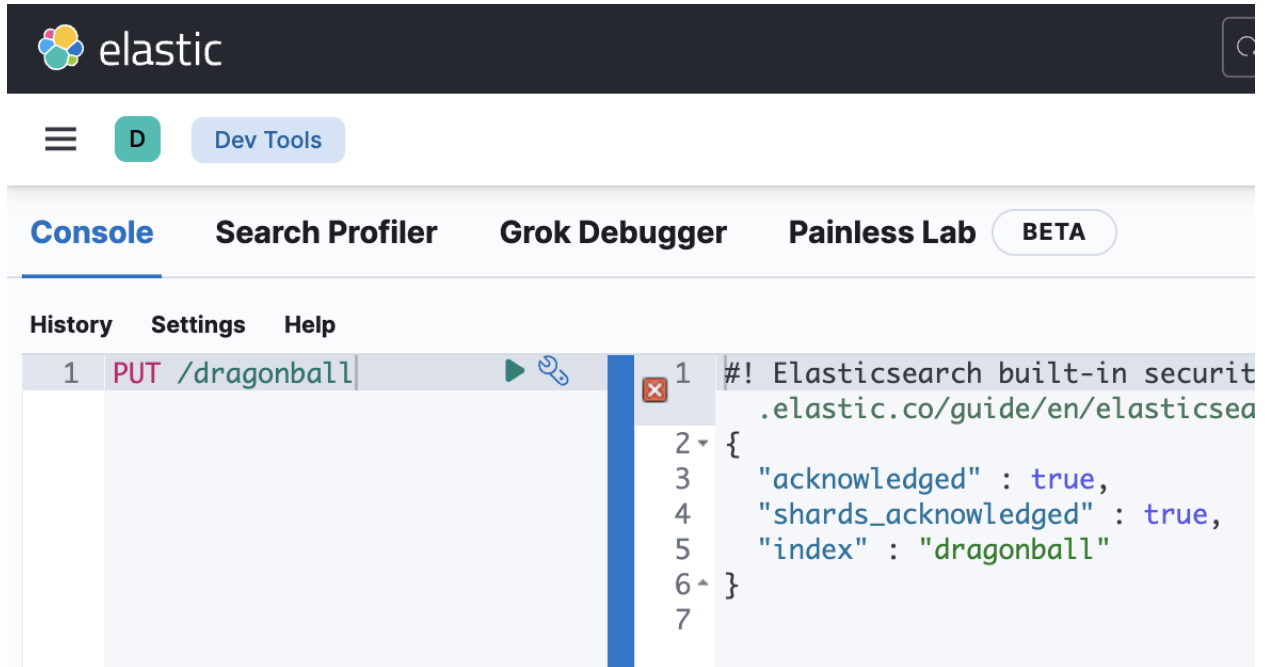


Creating a Resource

Now, let's try adding in data to our nosql database which in current case is ElasticSearch.

You can try creating a resource using the command

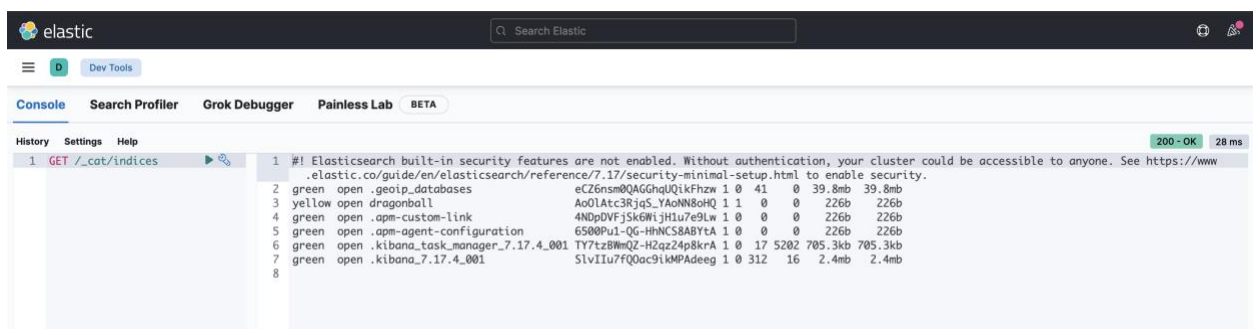
PUT /resourcename



Listing Indices

Now, try out the list indices call to observe any changes in the resources using,

GET /_cat/indices

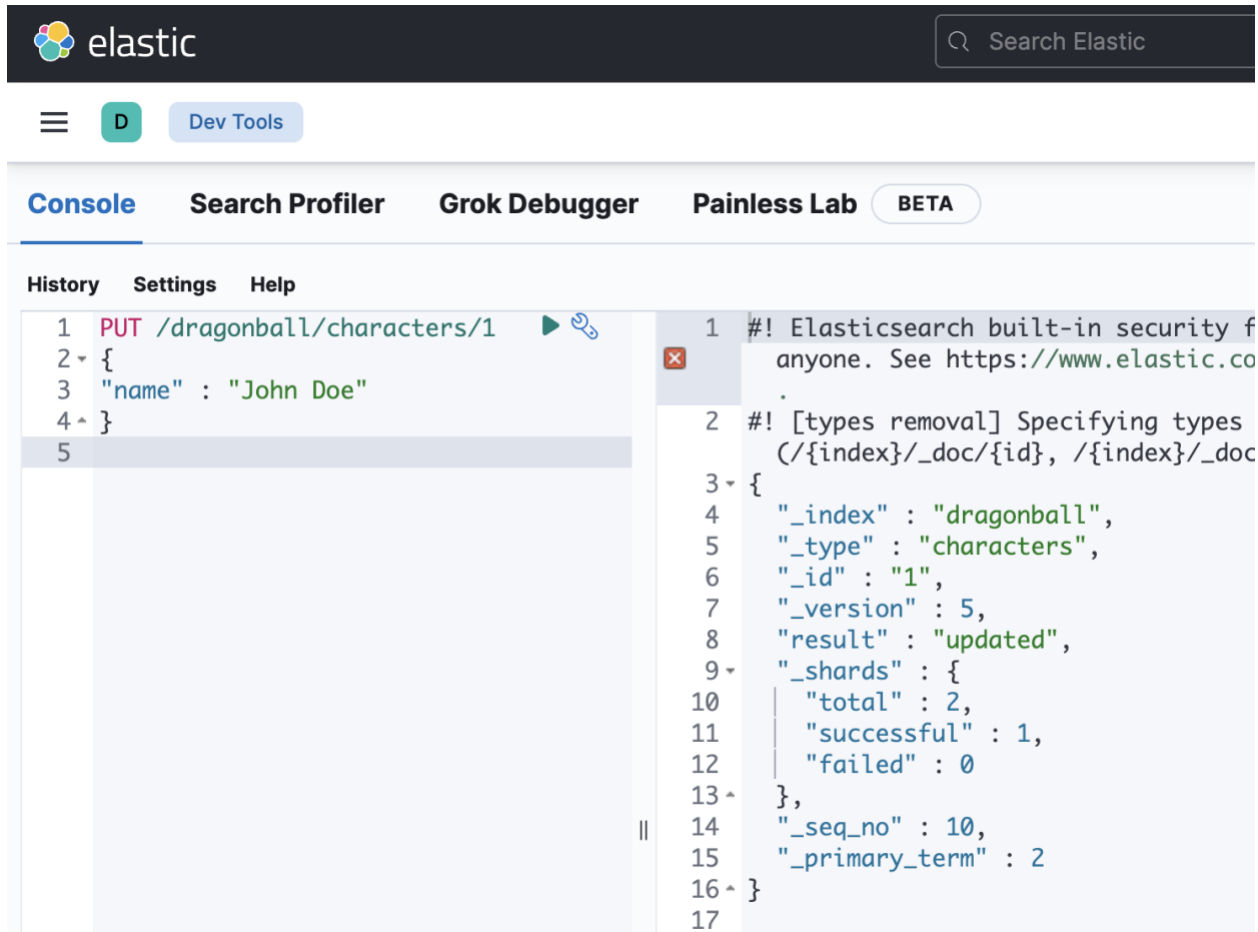


Next on, let's move ahead with addition of data into the resource we just created,

Adding Data to Resource

PUT /resourcename/colname/1

```
{  
  "name" : "John Doe",  
}
```



Note: I had already inserted four data in same document 1 before “John Doe”, thus is the reason for version 5.

Exercise To Do:

I helped you add John Doe, its your turn to add in your personal data (consider populating it with your name, class group and email) to the same resource.

The screenshot shows the Elastic Dev Tools interface. The top bar includes the Elastic logo, a search bar, and a 'Dev Tools' button. Below this is a navigation bar with 'Console', 'Search Profiler', 'Grok Debugger', 'Painless Lab', and a 'BETA' badge. The 'Console' tab is active, showing a 'History' section with a single entry. The main area displays a REST client request and response. The request is a PUT to `/dragonball/characters/5` with a JSON body containing name, class_group, and email. The response is a JSON object with metadata and the created character data.

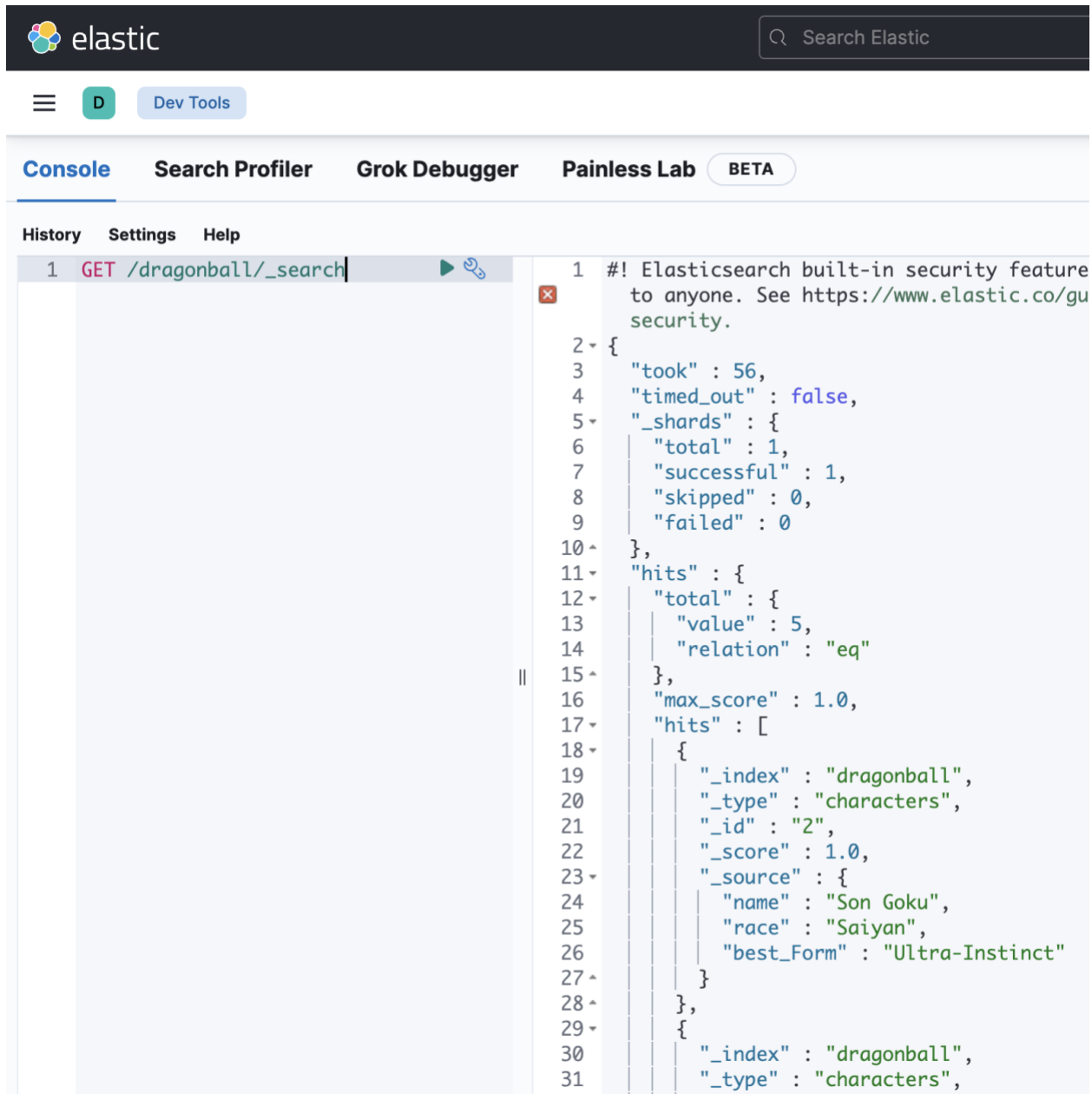
```
1 PUT /dragonball/characters/5
2 {
3   "name" : "Narayan Lohani",
4   "class_group" : "L6CG15",
5   "email": "this.is.narayan@gmail.com"
6 }
7
```

```
1 #! Elasticsearch built-in security
2 #! [types removal] Specifying type endpoints instead (/index/_doc)
3 {
4   "_index" : "dragonball",
5   "_type" : "characters",
6   "_id" : "5",
7   "_version" : 1,
8   "result" : "created",
9   "_shards" : {
10     "total" : 2,
11     "successful" : 1,
12     "failed" : 0
13   },
14   "_seq_no" : 9,
15   "_primary_term" : 2
16 }
```

Fetching the Record

Inorder to query you can try

GET /resourcename/_search



The screenshot shows the Elastic Dev Tools interface. At the top, the Elastic logo and a search bar are visible. Below the navigation bar, the 'Console' tab is selected. The console shows a single request: a GET request to /dragonball/_search. The response is a JSON object indicating a successful search with 5 hits. The first hit is for a character named Son Goku, who is a Saiyan and has the Ultra-Instinct form.

```
1 GET /dragonball/_search
```

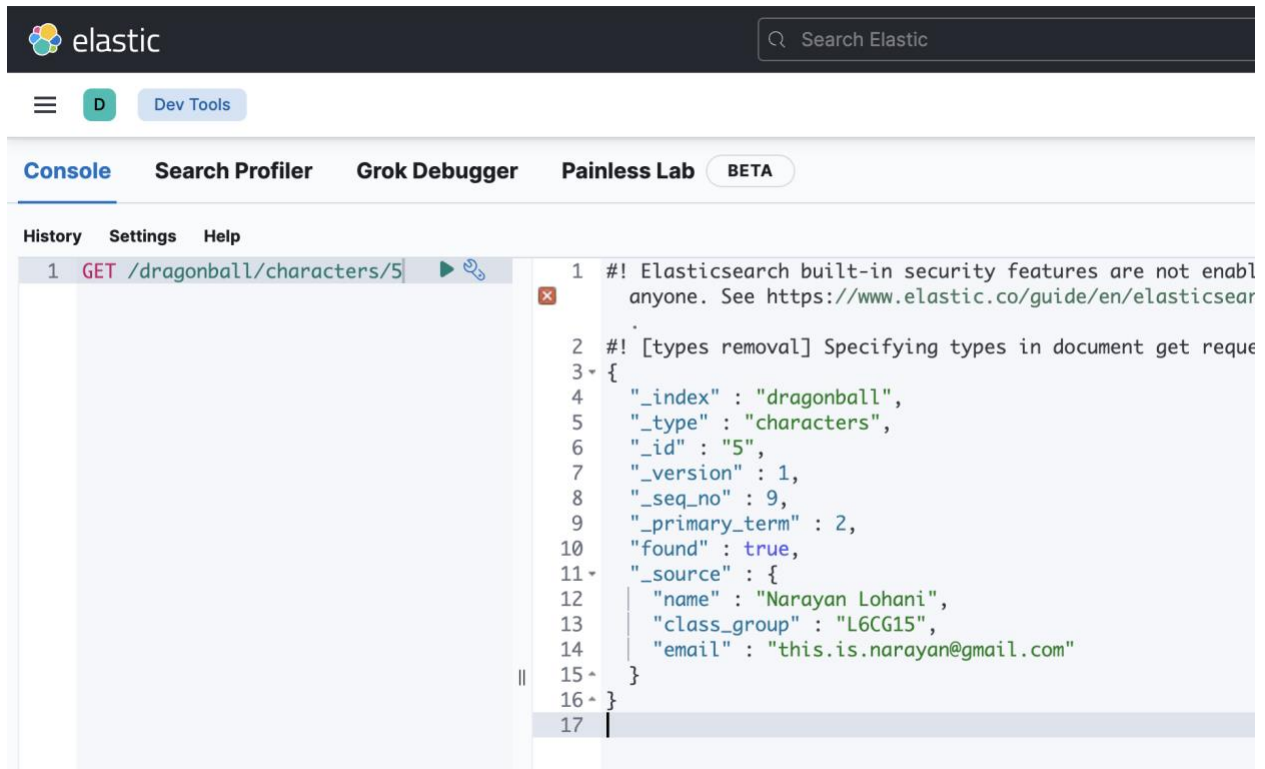
```
1  #! Elasticsearch built-in security feature
2  to anyone. See https://www.elastic.co/gu
3  security.
4  {
5    "took" : 56,
6    "timed_out" : false,
7    "_shards" : {
8      "total" : 1,
9      "successful" : 1,
10     "skipped" : 0,
11     "failed" : 0
12   },
13   "hits" : {
14     "total" : {
15       "value" : 5,
16       "relation" : "eq"
17     },
18     "max_score" : 1.0,
19     "hits" : [
20       {
21         "_index" : "dragonball",
22         "_type" : "characters",
23         "_id" : "2",
24         "_score" : 1.0,
25         "_source" : {
26           "name" : "Son Goku",
27           "race" : "Saiyan",
28           "best_Form" : "Ultra-Instinct"
29         }
30       },
31       {
32         "_index" : "dragonball",
33         "_type" : "characters",
```

Exercise to do

1. Write an api query call to fetch the second data record (that was your personal details).

Because I had inserted my data at `/dragonball/character/5`, the document id became 5.

Thus fetching the document with id 5 (which is my personal details).



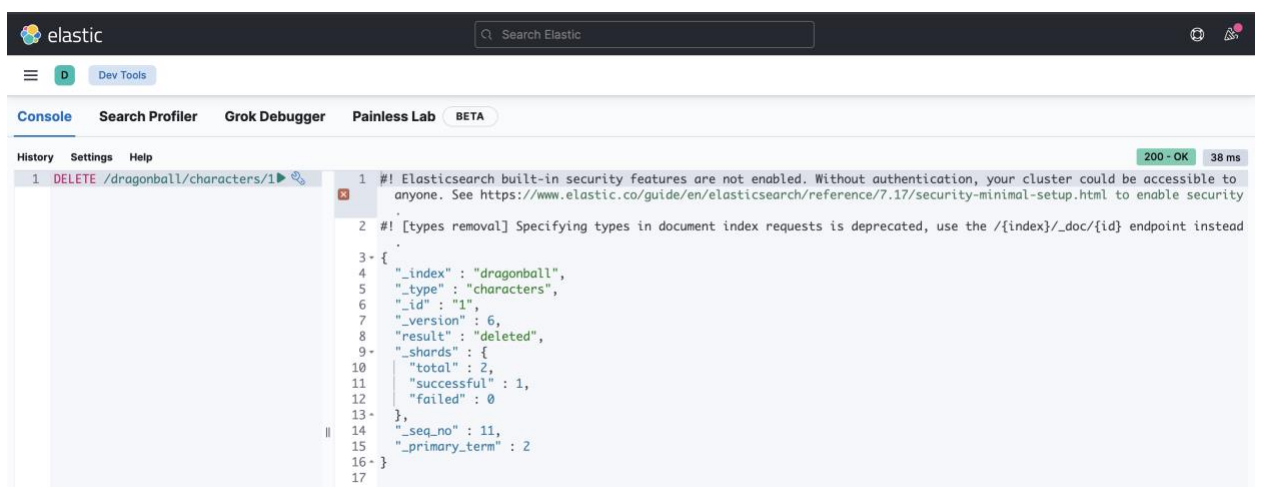
The screenshot shows the Elastic Dev Tools interface. The 'Console' tab is active. A GET request to `/dragonball/characters/5` has been executed. The response is a JSON object representing a character document.

```
1 GET /dragonball/characters/5
```

```
1 200 OK 38 ms
2 {
3   "_index": "dragonball",
4   "_type": "characters",
5   "_id": "5",
6   "_version": 1,
7   "_seq_no": 9,
8   "_primary_term": 2,
9   "found": true,
10  "_source": {
11    "name": "Narayan Lohani",
12    "class_group": "L6CG15",
13    "email": "this.is.narayan@gmail.com"
14  }
15 }
16 }
```

2. Write a query to remove the data record of John Doe from the resource.

John Doe data was inserted at `_id : 1`, so deleting that document will delete the record.



The screenshot shows the Elastic Dev Tools interface. The 'Console' tab is active. A DELETE request to `/dragonball/characters/1` has been executed. The response is a JSON object indicating the deletion was successful.

```
1 DELETE /dragonball/characters/1
```

```
1 200 OK 38 ms
2 {
3   "_index": "dragonball",
4   "_type": "characters",
5   "_id": "1",
6   "_version": 6,
7   "result": "deleted",
8   "_shards": {
9     "total": 2,
10    "successful": 1,
11    "failed": 0
12  },
13  "_seq_no": 11,
14  "_primary_term": 2
15 }
16 }
```